

HAIKANG DENG

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EDUCATION

University of California, Los Angeles

Sep 2024 - Present

Ph.D. in Computer Science, Artificial Intelligence

Advisor: Nanyun Peng

University of North Carolina, Chapel Hill

Aug 2019 - May 2023

B.S. in Computer Science & Statistics

GPA: 3.96/4.0

Advisor: Colin Raffel, Guanting Chen

PUBLICATIONS

- [1] Decoupling Task-Solving and Output Formatting in LLM Generation
Haikang Deng, Po-Nien Kung, Nanyun Peng
Annual Meeting of the Association for Computational Linguistics (ACL), 2026
- [2] Reward-Augmented Decoding: Efficient Controlled Text Generation With a Unidirectional Reward Model
Haikang Deng and Colin Raffel
Empirical Methods in Natural Language Processing (EMNLP), 2023
- [3] Large Language Models Struggle to Learn Long-Tail Knowledge
Nikhil Kandpal, **Haikang Deng**, Adam Roberts, Eric Wallace, and Colin Raffel
International Conference on Machine Learning (ICML), 2023
- [4] DialectGen: Benchmarking and Improving Dialect Robustness in Multimodal Generation
Yu Zhou, Sohyun An, **Haikang Deng**, Da Yin, Clark Peng, Cho-Jui Hsieh, Kai-Wei Chang, Nanyun Peng
ResponsibleFM Workshop @ NeurIPS, 2025

RESEARCH & PROFESSIONAL EXPERIENCE

University of North Carolina, Chapel Hill

Aug 2022 - Mar 2024

Research Assistant

Advised by Prof. Colin Raffel

- Conducted research on human feedback alignment, controlled text generation, and retrieval augmentation
- Trained expert model routing and tagging in a contrastive manner with signals from a reward model

Amazon

Dec 2023 - Jul 2024

Software Dev Engineer

Bellevue, WA

- Designed automation for product pre-compute based on given marketplace, merchant, and store utilization score
- Composed a product lead time update tool that supports bulk updates (up to 1 million entries)
- Built a Local Landing Page to display local products available for pick up

RESEARCH PROJECTS

Analysis of RLVR's Effect on Model Reasoning Priors

Sep 2025 - Present

- Analyze distinct reasoning patterns in model-generated solutions to mathematical problems
- Cluster responses using reasoning chain logic and semantic similarities to correlate reasoning strategies with solution outcomes
- Quantify evolution of sequence probability, pattern frequency, and reasoning chain composition throughout RL training

Power Prover Models with Curriculum Learning

Jul 2025 - Present

- Benchmark prover model's performance on Terence Tao's Analysis textbook exercise problems
- Design a curriculum learning framework for prover models using a mixture of SFT and RLVR
- Generate synthetic training question through conjecturing and generating instantiations

Neurally Decomposed Oracle for Preference Optimization

Jul 2024 - Nov 2024

- Derived training objectives from NADO and adapt to preference data
- Analyzed model-generated response as compared to SOTA methods such as DPO
- Interpolated and extrapolated policy model to explore the potential of expedited training

Embedding-Based Routing and Tagging of PEFT Modules

Oct 2023 - Mar 2024

- Trained a router that routes a query to a PEFT module using input embeddings
- Examined performance gap between held-in and held-out data routing
- Enabled zero-shot PEFT module addition to the pool of modules

Tracing Model-Generated Wrong Answers

Nov 2022 - Feb 2023

- Explored correlation between QA accuracy and the number of relevant documents that contain negative co-occurrences
- Identified that incorrect answers are not driven by an increased association between the wrong answer and query entities